

Problem Set 4

ECON 2010 – Macroeconomics

Spring 2018

Due at the start of class Week 10, Wednesday, March 21. You have **one week** to complete this problem set. No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct (some of) the problem set together in class.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (24 points)

President Trump has frequently discussed his desire to increase economic growth in the US. In particular, he's extolled reaching 4% real GDP growth. As he said in one 2017 speech: "They're saying three, but I say four." He's even aimed higher at times: "And I say there's no reason we shouldn't be able to get, at some point into the future, to five and above." But most often he talks about 4% as the magic number.

In 2017, real GDP growth in the US was 2.5%. You might not think 4 is that much larger than 2.5 — some countries like China have grown upwards of 6%, as high as 11% even, in recent decades — but it would take quite a lot for an advanced economy like the US to reach 4% from where we are today. Even to reach 3% would take a lot. To grow 3 percent a year for 10 years means that

in a decade, the U.S. economy would be a third larger (would be producing a third more) than it is now! Given how much we *already* produce, that's a lot.

Each of the following quotations come from a real NPR radio show (Planet Money) in which an economist discusses exactly *what* it would take for the US to increase its real GDP growth rate from 2.5% to 4%. For each quotation, indicate **which factor of production** the quotation is referring to. Your choices for factors of production are **labor, capital, human capital, or technology/entrepreneurship**. Choose just **one** factor of production for each quotation.

- (a) "So we would need two dot-com booms going simultaneously, at the same time." **(3 points)**

- (b) "What if we had... some better education and training?" **(3 points)**

- (c) "We would basically need to double our immigrant population over the next decade." **(3 points)**

- (d) "We have a lot of signals we send people to tell them to retire early. The Social Security early age is 62. The normal age is 66 and not really going to rise with life expectancy. The Medicare age is 65. You can pull out from your IRA at 59 and a half. We can create tax breaks, like a payroll tax break, for folks that work longer. We could change a lot of the regulations and rules to encourage employers to keep people on for longer." **(3 points)**

- (e) "...more government building projects, infrastructure, more objects, tools... Hammers, microphones, speakers – I use Microsoft Excel. That's one of my best pieces of [equipment] in my job." **(3 points)**

- (f) "What you'd also have to do is take every single adult age 25 to 55, you know, every single prime-age adult and force them to work. And that includes folks that are home because they're taking care of their kids. That includes folks that are in jail, that are on disability. And it includes some folks that legitimately could be working with the right mix of incentives and support." **(3 points)**

(g) “If we were to [develop something like the desktop computer] again, if we were – if something were invented that every single office would absolutely have to buy, whether it’s, I don’t know, virtual reality or hologram, something that just transforms business in a way and people have to pour money into it...” **(3 points)**

(h) “We could just all get a whole lot better at our jobs.” **(3 points)**

Problem 2 (14 points)

These should all be in your notes, and they’re in the book, but... confusion persists.

Simplifying assumption: An assumption that reduces the complexity of an economic problem. Simplifying assumptions might reduce the number of things (i.e. people, firms, households) being studied, the number of places being studied, the time period of study, and so forth.

Assumption of rational actors: Actors are trying to maximize their utility (i.e. their well-being); often assumed to be maximizing their *material* well-being, like with money.

Normative versus positive: Normative statements are those about how the world *should* or *ought* to work. They are subjective statements. Positive statements are statements about how the world *does* work or how the world *is*. Positive statements are not necessarily *true*. But they are objective statements. It’s possible to know whether a positive statement is true or not. It’s not possible to know if a normative statement is true or not.

Equilibrium: A situation in which no one has an incentive to change their behavior given the behavior of others.

Real versus nominal: Real figures (numbers, prices) take into account changing price levels (i.e. inflation), while nominal figures (numbers, prices) are measured with current prices (in whatever year the figure is being calculated). Real figures allow you to compare figures (numbers, prices) across time without differences being attributed to price levels (i.e. inflation). Nominal figures (numbers, prices) might be different across time simply because of differences in price levels (i.e. inflation).

All else fixed (ceteris parabis): Considering only one change at a time. Allows you to consider the effect/impact of just *one* variable/factor on another. All else fixed assumptions are about

change as opposed to simplifying assumptions which are about the overall scope of a problem.

Thinking on the margin: Consideration of a small/incremental change, usually a single unit (whatever the relevant unit in the problem may be; i.e. \$1, 1 hour, 1 minute, 1 person, and so on).

Opportunity cost: The cost of doing something in terms of resources given up. Put differently, an opportunity cost represents an alternative given up when a choice is made. Resources that are given up are not just money. Time is a resource frequently given up in order to do something.

For each of the quotations below, **circle** which of the two available choices better reflects the idea *explicitly* referenced in the quotation.

- (a) “It seems like economists are never taking a stand on any policy — they just tell you what they think the policy will do, but they don’t tell you if you should do it.” **(2 points)**
(A) Distinguishing real from nominal (B) Distinguishing normative from positive
- (b) “When you drive just one more minute on the road, you cost all commuters a combined \$4 million.” **(2 points)**
(A) Thinking on the margin (B) Rational actors
- (c) “After I’ve already eaten most of a bag of potato chips, I find that I don’t really enjoy that last potato chip as much as the one before it, and certainly less than the first.” **(2 points)**
(A) Equilibrium (B) Thinking on the margin
- (d) “The market demand curve for bananas indicates that a change in the price from \$10 to \$11 results in a significant drop in quantity demanded.” **(2 points)**
(A) Distinguishing real from nominal (B) Thinking on the margin

(e) “I want to know what happens when U.S. government spending goes up, but let’s leave aside whether anything changes in terms of China’s imports from or exports to the U.S.” **(2 points)**

(A) Simplifying assumption (B) All else fixed

(f) “Let’s consider what happens when tariffs (import taxes) on steel are increased as President Trump has proposed. But let’s just consider the effects for the US steel industry, and the US auto industry.” **(2 points)**

(A) Simplifying assumption (B) All else fixed

(g) “Assume that people want to maximize their leisure time and their incomes and they don’t care about anything else.” **(2 points)**

(A) Assumption of rational actors (B) Simplifying assumption

Problem 3 (27 points)

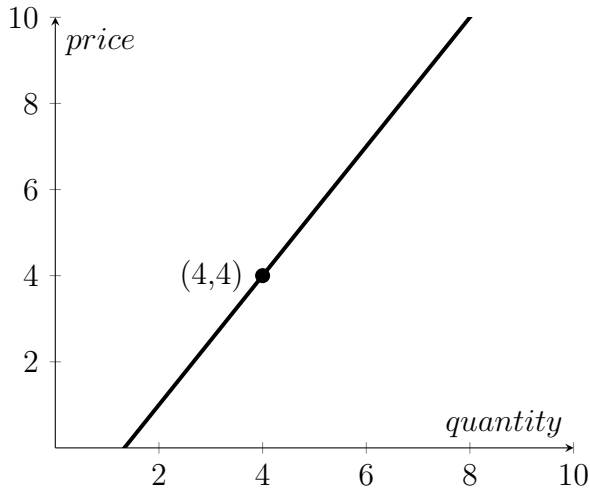
Consider the market for dog bones for pets in the United States. Let’s assume that there are many manufacturers (firms who produce) of dog bones. And because this is just one market, dog bones are indistinguishable across manufacturers. To make this market seem more like the real world, we won’t even put a number on how many manufacturers there are.

Even though dogs eventually consume the dog bones, it’s human beings (dog owners) in the US who demand dog bones. The number of individual consumers is quite large — on the order of millions of people, but we won’t say here exactly how many, again, to make this more like a real-world market. We know that there are a lot of individuals who make up demand, but we don’t always know exactly how many across an entire market, especially a large one like for dog bones.)

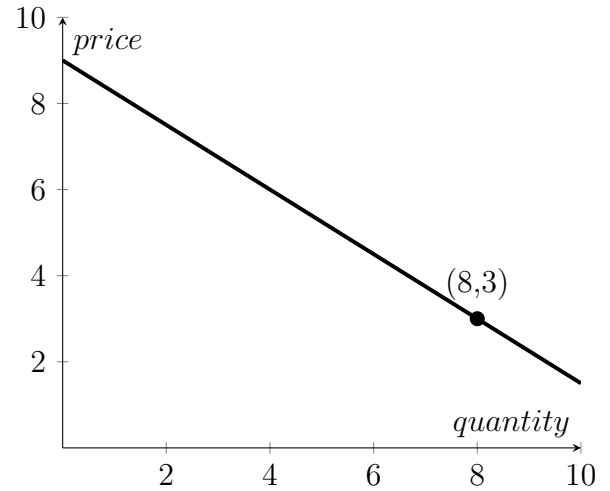
In Figure 1, I’ve drawn the market supply curve for dog bones in Plot A, the market demand curve for dog bones in Plot B, and the market equilibrium that puts the market demand and market supply curves together on one plot (Plot C). The point at which the two curves cross in Plot C is *approximately* (5,5). (It’s not quite there, but that’s close enough for our purposes.)

Figure 1: Market supply, market demand and equilibrium for dog bones

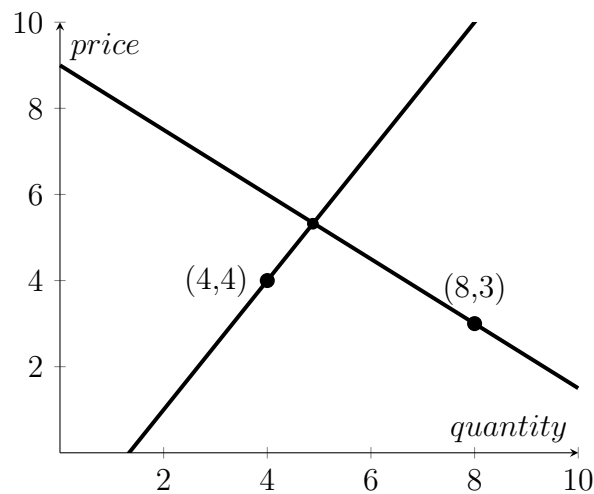
Plot A: Market supply curve



Plot B: Market demand curve



Plot C: Market supply and market demand, a.k.a. market equilibrium



NOTE: The price in all of the plots is measured in *dollars* per unit (i.e. price per dog bone). The quantity in all of the plots is measured in *millions of dog bones*. So, a 2 on the quantity axis represents *2 million dog bones*.

Given this information about the market, answer the following questions.

- (a) In **one or at most two complete sentences**, state what is represented by the point (4,4) in Plot A (the market supply curve). Be thorough in your answer. To receive full credit you cannot simply indicate how each number is measured (i.e. you can't just say dollars for one of the numbers), you need to state what the numbers together *mean conceptually*. **(2 points)**

- (b) In **one or at most two complete sentences**, state what is represented by the point (8,3) in Plot B (the market demand curve). Be thorough in your answer. To receive full credit you cannot simply indicate how each number is measured (i.e. you can't just say dollars for one of the numbers), you need to state what the numbers together *mean conceptually*. **(2 points)**
- (c) Given this market equilibrium in Plot C, explain briefly why a consumer who wants to buy a dog bone at \$4 (i.e. who demands a dog bone at \$4) cannot buy a dog bone at that price. Hold all else fixed when you answer this question. **(4 points)**
- (d) Given the market equilibrium in Plot C, explain briefly why a firm/producer who wants to sell a dog bone at \$6 (i.e. who supplies a dog bone at \$6) cannot sell a dog bone at that price. Hold all else fixed when you answer this question. **(4 points)**

(e) Given the market equilibrium in Plot C, consider the following change in the market. Consider that a million new people move to the US. Most of these people are dog owners who feed their dogs bones. Answer the following questions about this change to the market. Hold all else fixed when you consider this change to the market. **(15 points total)**

(1) Which curve or curves will shift as a result of this market change? You could say **market supply**, **market demand**, **none** or **both**. **(2 points)**

(2) In which direction will the curve or curves shift that you indicated would shift in part (e1)? For each you can say **right** or **left**. If no curves shift, just write “none” here. **(2 points)**

(3) In just a few sentences, explain briefly **why** the curve or curves shift in the direction that you indicated they would shift. If no curves shift, just write “none” here. **(5 points)**

- (4) Draw a new plot that indicates the curve shift or shifts that you thought would take place. **Draw** the starting curves and equilibrium from Plot C with solid lines, and **draw** any curve shifts with dashed lines. **Indicate** the starting equilibrium from Plot C with a point and label it **A**. **Indicate** the new equilibrium according to your curve shifts with a point and label it **B**. If no curves shift, just re-draw Plot C.

(This is not a drawing test; just draw the shifted curves well enough to visualize any equilibrium change that may have occurred. I do not give you enough information to know by how much the curves shifted, if at all.) **(3 points)**

- (5) According to the new equilibrium, how have **each** price and quantity changed (if at all). For each price and quantity you can say that they have gone **up**, **down**, **stayed the same**, or the change is **not possible to know** given the information provided. (Remember: An equilibrium is a price **and** a quantity. You cannot correctly answer this question by just saying “down” or “up.” An equilibrium can’t go “down” or “up.” *What* went down; *what* went up?) **(3 points)**

Problem 4 (27 points)

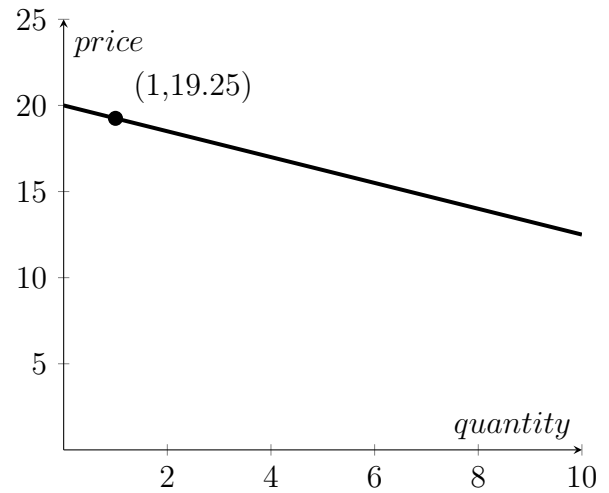
Consider the labor market for drywall installers in the United States. As we did in Problem 3 above, let’s assume that we don’t know exactly how many firms there are in the US that hire drywall installers or exactly how many individual laborers there are in the market. We’ll just say that there are many many firms, and many many individual laborers. Assume that all drywall installers are indistinguishable to firms, and all drywall firms are indistinguishable to drywall installers (this is an example of a simplifying assumption, but it’s also an assumption that’s part of our definition of a market).

Figure 2: Market supply, market demand and equilibrium for drywall installers

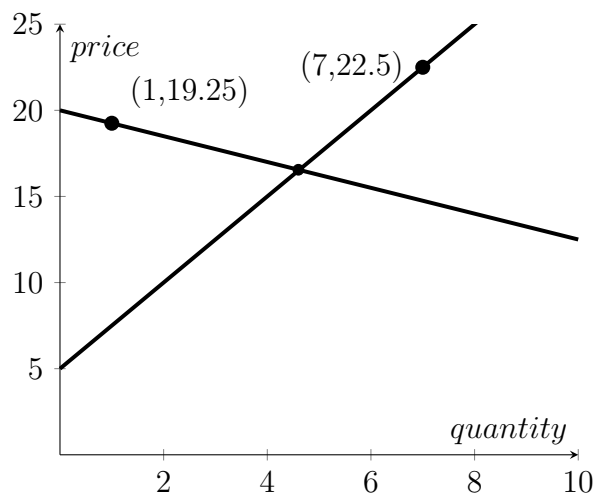
Plot A: Market supply curve



Plot B: Market demand curve



Plot C: Market supply and market demand, a.k.a. market equilibrium



In Figure 1, I've drawn the labor market supply curve for drywall installers in Plot A, the labor market demand curve for drywall installers in Plot B, and the market equilibrium that puts the market demand and market supply curves together on one plot (Plot C). The point at which the two curves cross in Plot C is *approximately* $(4.6, 16.5)$. (It's not quite there, but that's close enough for our purposes.)

NOTE: The price (i.e. the wage) in all of the plots is measured in *dollars per hour* of work (i.e. it's an hourly wage). The quantity in all of the plots is measured in *millions of total hours* worked. So, a 2 on the quantity axis represents 2 million total hours worked.

As you answer this question, assume (simplifying assumption) that all workers and all firms are

located in exactly the same place. That is, don't try to complicate the problem by thinking about where workers are located geographically.

Given this information about the market, answer the following questions.

- (a) In **one or at most two complete sentences**, state what is represented by the point (7,22.5) in Plot A (the labor market supply curve). Be thorough in your answer. To receive full credit you cannot simply indicate how each number is measured (i.e. you can't just say dollars for one of the numbers), you need to state what the numbers together *mean conceptually*. **(2 points)**
- (b) In **one or at most two complete sentences**, state what is represented by the point (1,19.25) in Plot B (the labor market demand curve). Be thorough in your answer. To receive full credit you cannot simply indicate how each number is measured (i.e. you can't just say dollars for one of the numbers), you need to state what the numbers together *mean conceptually*. **(2 points)**
- (c) Given this market equilibrium in Plot C, explain briefly why a consumer who wants to earn a wage of \$18 per hour (i.e. who supplies their labor at the hourly wage of \$18) cannot find hours to work at that wage. Hold all else fixed when you answer this question. **(4 points)**

- (d) Given the market equilibrium in Plot C, explain briefly why a firm/producer who wants to hire a drywall installer at \$15 (i.e. who demands drywall installation at \$15) cannot buy labor at that price. Hold all else fixed when you answer this question. **(4 points)**

- (e) Given the market equilibrium in Plot C, consider the following change in the market. **First**, consider an increase in US immigration. This new population needs houses, which have drywall in them. Therefore the price that drywall installation firms can charge homebuilders to install drywall goes up. (Note: This market change is **not in and of itself** a change in the wage for a drywall installer. It is a change in the price of the *output* that drywall installation *firms* provide. Drywall installers are just *one* input for drywall installation firms. So, I'm not saying that the market change is a change in the wage; I'm saying that something we usually hold fixed when we consider changes in wages — i.e. the price of the output product for which the labor we're considering is an input — has changed.) Assume that *none* of the new immigrants can install drywall.

Second, in addition to the first change, consider a massive shortage of nail guns in the US. Nail guns are an important tool used by drywall installers to install drywall. Without nail guns, to install the same amount of drywall as before requires drywall installers to hammer nails in by hand, which takes much longer. Assume drywall installers have no preference about whether they use a nail gun or hammer nails. That is, they are completely indifferent about which tool they use.

Answer the following questions about this change to the market. Hold all else fixed when you consider these change to the market. **(15 points total)**

- (1) Which curve or curves will shift as a result of these market changes? You could say **labor market supply, labor market demand, none or both**. **(2 points)**

- (2) In which direction will the curve or curves shift that you indicated would shift in part (e1)? For each you can say **right** or **left**. If no curves shift, just write “none” here. **(2 points)**
- (3) In just a few sentences, explain briefly **why** the curve or curves shift in the direction that you indicated they would shift. If no curves shift, just write “none” here. **(5 points)**
- (4) Draw a new plot that indicates the curve shift or shifts that you thought would take place. **Draw** the starting curves and equilibrium from Plot C with solid lines, and **draw** any curve shifts with dashed lines. **Indicate** the starting equilibrium from Plot C with a point and label it **A**. **Indicate** the new equilibrium according to your curve shifts with a point and label it **B**. If no curves shift, just re-draw Plot C.
- (This is not a drawing test; just draw the shifted curves well enough to visualize any equilibrium change that may have occurred. I do not give you enough information to know by how much the curves shifted, if at all.) **(3 points)**

- (5) According to the new equilibrium, how have **each** price (i.e. the wage) and quantity changed (if at all). For each price and quantity you can say that they have gone **up**, **down**, **stayed the same**, or the change is **not possible to know** given the information provided. (Remember: An equilibrium is a price **and** a quantity. You cannot correctly answer this question by just saying “down” or “up.” An equilibrium can’t go “down” or “up.” *What* went down; *what* went up?) **(3 points)**

Correcting Your Problem Set (8 points)

You’ll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.